

Nocturnal blood pressure is elevated with natriuresis and proteinuria as renal function deteriorates in nephropathy

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Nocturnal blood pressure is elevated with natriuresis and proteinuria as renal function deteriorates in nephropathy.

Background. We reported that patients with sodium sensitive type of hypertension exhibited the lack of nocturnal fall in blood pressure with enhanced natriuresis during night. Sodium sensitivity is caused by diminished glomerular filtration capability and/or augmented tubular reabsorption of sodium, and seems tightly linked with glomerular capillary hypertension. In the present study, we investigated the relationship between glomerular filtration rate and circadian rhythms of these parameters in patients with glomerulopathy.

Methods. Twenty six patients (15 men and 11 women; aged 17 to 72 years; mean age 47 ± 3 years), whose diagnosis was confirmed as glomerulopathy with renal biopsy, were studied during hospitalization. Ambulatory blood pressure for 24 hours was monitored, while urinary samples were collected for both daytime (6:00 a.m. to 9:00 p.m.) and nighttime (9:00 p.m. to 6:00 a.m.) to estimate circadian rhythms of urinary sodium and protein excretion rates ($U_{Na}V$, $U_{pro}V$). Then night/day ratios of mean arterial blood pressure (MAP), $U_{Na}V$, and $U_{pro}V$ were analyzed in relation to 24-hour creatinine clearance as a marker of glomerular filtration rate.

Results. Serum creatinine and creatinine clearance were 1.1 ± 0.1 mg/dL and 89 ± 7 mL/min/1.73 m². There were significant day-night differences in MAP (96 ± 2 mm Hg vs. 92 ± 2 mm Hg; $P = 0.006$), $U_{Na}V$ (6.7 ± 0.9 mmol/hour vs. 3.6 ± 0.3 mmol/hour; $P = 0.003$), and $U_{pro}V$ (161 ± 27 mg/hour vs. 128 ± 28 mg/hour; $P = 0.02$). Creatinine clearance had significantly negative relationships with night/day ratios of MAP ($r = -0.49$; $P = 0.01$), $U_{Na}V$ ($r = -0.43$; $P = 0.03$), and $U_{pro}V$ ($r = -0.41$; $P = 0.04$). In addition, night/day ratio of MAP had significantly positive relationships with night/day ratios of $U_{Na}V$ ($r = 0.49$; $P = 0.01$) and $U_{pro}V$ ($r = 0.45$; $P = 0.02$).

Conclusion. Our results show that as renal function deteriorates in glomerulopathy the nocturnal dip in blood pressure is lost, resulting in enhanced urinary sodium and protein excretions during night. These findings are compatible with our proposal that impaired natriuresis during daytime makes noc-

turnal blood pressure elevated to compensate for diminished natriuresis by pressure natriuresis. We speculate that nocturnal glomerular capillary hypertension contributes, at least in part, to enhanced urinary sodium and protein excretions during night.

In some patients with hypertension [1–6] or renal dysfunction [7–10], blood pressure fails to fall during night and they have been called “nondipper,” whereas those with a normal nocturnal blood pressure dip are called “dipper.” Many investigators have reported that nondippers were certainly exposed to greater risks of cardiovascular complications such as left ventricular hypertrophy, cerebrovascular disease, microalbuminuria, and end-stage renal disease (ESRD) than dippers [1–3, 10–16]. We showed that the circadian rhythms of natriuresis was disturbed in patients with non-dipper type of essential hypertension [17], whose blood pressure was usually sodium sensitive [4, 5]. Furthermore, in sodium-sensitive type of essential hypertension, we have proposed that glomerular capillary pressure is elevated [18–20], and urinary albumin excretion rate, which may be a marker of glomerular capillary pressure [21, 22], is reported to be greater than in the nonsodium-sensitive type [20, 22]. Since sodium sensitivity of blood pressure is considered to be caused by diminished glomerular filtration capability and/or augmented tubular reabsorption of sodium [18, 19], such disarrangement may be clearly recognized in glomerulopathy as renal function deteriorates. Therefore, we investigated the relationship between glomerular filtration rate (GFR) and circadian rhythms of these parameters as well as between circadian rhythms of blood pressure and those of urinary sodium and protein excretions in patients with glomerulopathy.

Key words: circadian rhythm, blood pressure, natriuresis, proteinuria, renal function.

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METHODS

Patients and study protocol

Twenty six patients with glomerulopathy (15 men and 11 women; aged 17 to 72 years with mean of 47 ± 3 years; body mass index 23.4 ± 0.8 kg/m²), who were admitted

Table 1. Patient characteristics of glomerulopathy

	All (N = 26)	Higher creatinine clearance (106–151 mL/min) (N = 9)	P value	Lower creatinine clearance (16–62 mL/min) (N = 9)
Age years	46.9 ± 3.1	41.6 ± 3.4	0.6	45.9 ± 6.9
Male/female	15/11	6/3	0.3	5/4
Body mass index kg/m ²	23.4 ± 0.8	24.0 ± 1.5	0.7	23.2 ± 1.6
Serum creatinine mg/dL	1.1 ± 0.1	0.9 ± 0.1	0.003	1.5 ± 0.2
Creatinine clearance mL/min/1.73 m ²	89 ± 7	126 ± 5	<0.0001	46 ± 6
Proteinuria g/day	3.6 ± 0.6	3.8 ± 1.3	0.4	2.6 ± 0.6
Systolic blood pressure mm Hg	127 ± 4	120 ± 5	0.04	137 ± 7
Diastolic blood pressure mm Hg	78 ± 2	81 ± 5	0.5	77 ± 3
Mean arterial pressure mm Hg	95 ± 2	94 ± 5	0.5	97 ± 3
Night/day ratio of mean arterial pressure	0.96 ± 0.02	0.91 ± 0.01	0.018	1.00 ± 0.03
Night/day ratio of U _{cr} V	1.01 ± 0.07	0.89 ± 0.10	0.5	0.99 ± 0.11
Night/day ratio of U _{Na} V	0.74 ± 0.09	0.46 ± 0.11	0.012	0.99 ± 0.13
Night/day ratio of U _{pro} V	0.93 ± 0.02	0.87 ± 0.04	0.012	0.98 ± 0.03

Values are mean ± SEM or number of patients. Abbreviations are: U_{cr}V, creatinine excretion rate; U_{Na}V, urinary sodium excretion rate; U_{pro}V, urinary protein excretion rate.

for renal biopsy because of unexplained deterioration in renal function or proteinuria more than 0.5 g/day, were studied in Nagoya City University Hospital during hospitalization following renal biopsy. All of the participants were consecutively enrolled after giving informed consent. When corticosteroids or immunosuppressive agents had been indicated, they were continuously given in the same doses in eight patients until the study ended. Patients ate a relatively low-sodium diet containing approximately 8 g/day of NaCl, and were allowed usual daily life activity, except they were asked to get up on 6:00 a.m. and to go to sleep on 9:00 p.m. Ambulatory blood pressure for 24 hours was monitored every 30 minutes noninvasively with a validated automatic device (model ES-H531) (Terumo, Tokyo, Japan), while urinary samples were collected for both daytime (6:00 a.m. to 9:00 p.m.) and nighttime (9:00 p.m. to 6:00 a.m.) to estimate circadian rhythms of urinary creatinine, sodium and protein excretion rates (U_{cr}V, U_{Na}V, U_{pro}V) as well as of urinary volume (V). Mean arterial pressure (MAP) was calculated as diastolic blood pressure plus one third of the pulse blood pressure. Daytime blood pressure was calculated as the average of the 30 readings between 6:00 a.m. and 9:00 p.m., whereas nighttime blood pressure was the average of the remaining 18 readings. Then, night/day ratios of MAP, U_{Na}V, and U_{pro}V were analyzed in relation to 24-hour creatinine clearance as a marker of GFR. Participants (N = 26) were divided into three groups, one third each, based on creatinine clearance: lower (16 to 62 mL/min; N = 9), medium (75 to 92 mL/min; N = 8), and higher (106 to 151 mL/min; N = 9) to compare the night/day ratios between higher and lower creatinine clearance groups.

Statistical analysis

Results are expressed as the mean ± SEM. The significance of differences between daytime and nighttime val-

ues was determined by Student *t* test for paired samples. The significance of differences among three groups with different level of GFR was tested by one-way analysis of variance (ANOVA), followed by post hoc Fisher's test to compare between higher and lower groups. The significance of differences in sex distribution between the two groups was examined by χ^2 test. The correlations among creatinine clearance and night/day ratios of MAP, U_{Na}V, and U_{pro}V were obtained by the method of least-squares. Logarithmic transformation was applied for proteinuria in the regression analysis.

RESULTS

We studied 26 patients with glomerulopathy, characteristics of whom are listed in Table 1. Their histologic diagnoses were poststreptococcal glomerulonephritis in one, IgA nephropathy in nine, membranous nephropathy in four, focal segmental glomerulosclerosis in two, lupus nephritis in eight, myeloperoxidase antineutrophil cytoplasmic antibody related vasculitis in one, and minor glomerular abnormalities in one. No patient with diabetic nephropathy was included. Proteinuria was greater than 0.7 g/day in all patients, and ranged between 0.7 and 11.8 g/day.

There were significant decreases from day to night in blood pressure and urinary excretion rates of sodium and protein, while no difference was found in creatinine excretion rate between day and night (Table 2). Creatinine clearance (mL/min) had significantly negative relationships with night/day ratios of MAP ($r = -0.49$; $P = 0.01$), U_{Na}V ($r = -0.43$; $P = 0.03$), and U_{pro}V ($r = -0.41$; $P = 0.04$ in the ratio of log proteinuria) as shown in Figure 1. However, creatinine clearance was not correlated with night/day ratio of U_{cr}V ($r = -0.05$, $P = 0.8$). As shown in Table 1, compared to patients with higher creatinine clearance (106 to 151 mL/min; N = 9), those

Table 2. Day/night blood pressure and other parameters in glomerulopathy

	Daytime (6:00 a.m. to 9:00 p.m.)	<i>P</i> value	Nighttime (9:00 p.m. to 6:00 a.m.)
Systolic blood pressure <i>mm Hg</i>	129 ± 3	0.02	124 ± 4
Diastolic blood pressure <i>mm Hg</i>	80 ± 2	0.004	76 ± 2
Mean arterial pressure <i>mm Hg</i>	96 ± 2	0.006	92 ± 2
Heart rate <i>beat/min</i>	74 ± 2	<0.0001	64 ± 8
V <i>mL/hour</i>	88 ± 8	0.001	64 ± 6
U _{cr} V <i>mg/hour</i>	53 ± 5	0.3	49 ± 4
U _{Na} V <i>mmol/hour</i>	6.7 ± 0.9	0.003	3.6 ± 0.3
U _{pro} V <i>mg/hour</i>	161 ± 27	0.02	128 ± 28

Mean ± SEM (*N* = 26). Abbreviations are: V, urinary volume; U_{cr}V, creatinine excretion rate; U_{Na}V, urinary sodium excretion rate; U_{pro}V, urinary protein excretion rate.

with lower creatinine clearance (16 to 62 mL/min; *N* = 9), had significantly higher night/day ratios of MAP, U_{Na}V, and U_{pro}V, but night/day ratio of U_{cr}V was not different between two groups.

In addition, night/day ratio of MAP had significantly positive relationships with night/day ratios of U_{Na}V (*r* = 0.49; *P* = 0.01) and U_{pro}V (*r* = 0.45; *P* = 0.02 in the ratio of log proteinuria) as shown in Figure 2. Night/day ratio of creatinine excretion rate was not correlated with night/day ratio of MAP (*r* = 0.05, *P* = 0.8). Creatinine clearance was not correlated with the absolute values of MAP in both daytime and nighttime (*r* = 0.007; *P* = 0.97, *r* = -0.29; *P* = 0.2, respectively). During either daytime or nighttime, none of U_{Na}V (daytime *r* = 0.09, *P* = 0.7; nighttime *r* = 0.09, *P* = 0.7) or U_{pro}V (daytime: *r* = 0.11, *P* = 0.6; night-time: *r* = 0.16, *P* = 0.4) was correlated with MAP in each duration.

DISCUSSION

The present study clearly demonstrated that as renal function deteriorated in patients with glomerulopathy the nocturnal blood pressure was elevated with enhanced urinary excretion rates of sodium and protein during night. Especially, our study is the first, to our knowledge, to show the nocturnal natriuresis and proteinuria were enhanced with nocturnal hypertension as renal function deteriorated.

We previously reported in patients with essential hypertension that as blood pressure became more sodium sensitive, nocturnal decline in blood pressure was diminished [4]. Since glomerular filtration capability is one of the major factors determining sodium sensitivity [18, 19], our present findings that as creatinine clearance, a marker of GFR, is reduced the nocturnal decline in blood pressure is less pronounced seem reasonable. As we previ-

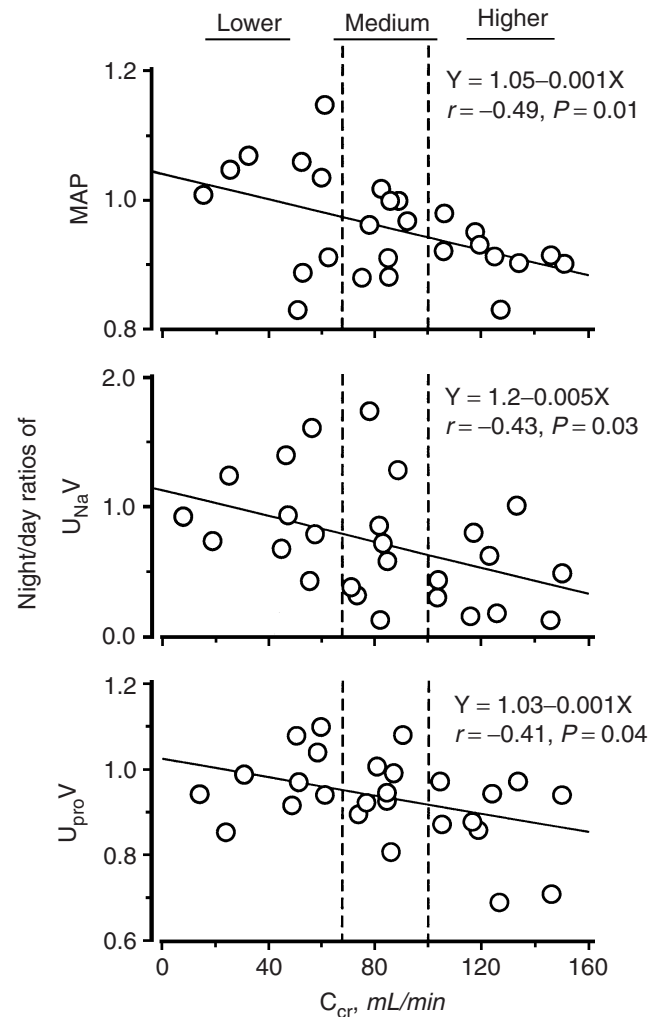


Fig. 1. Relationship of creatinine clearance (Ccr) (mL/min) as a marker of renal function with night/day ratios of mean arterial pressure (MAP) (mm Hg), urinary sodium excretion rate (U_{Na}V) (mmol/hour), and proteinuria (U_{pro}V) (mg/hour) in glomerulopathy (*N* = 26). Night/day ratio of proteinuria was calculated after logarithmic transformation. Night/day ratio of proteinuria itself without logarithmic transformation also tended to be negatively correlated with creatinine clearance (*r* = -0.35, *P* = 0.08). Patients were divided into three groups, one third each, based on creatinine clearance: lower (16 to 62 mL/min; *N* = 9), medium (75 to 92 mL/min; *N* = 8), and higher (106 to 151 mL/min; *N* = 9) to compare the night/day ratios between higher and lower creatinine clearance groups (see Table 1).

ously proposed for nondipping mechanisms in sodium-sensitive hypertension [4, 5, 17, 23], our present findings may further implicate that the diminished sodium excretory capability in glomerulopathy determines the circadian rhythm of blood pressure. Defect in sodium excretory capability, which becomes more evident as renal function deteriorates, makes MAP during the night elevated (that is nocturnal hypertension or nondipper), to compensate for diminished natriuresis during the day and to enhance pressure natriuresis during night. In fact, there was a significantly positive relationship between

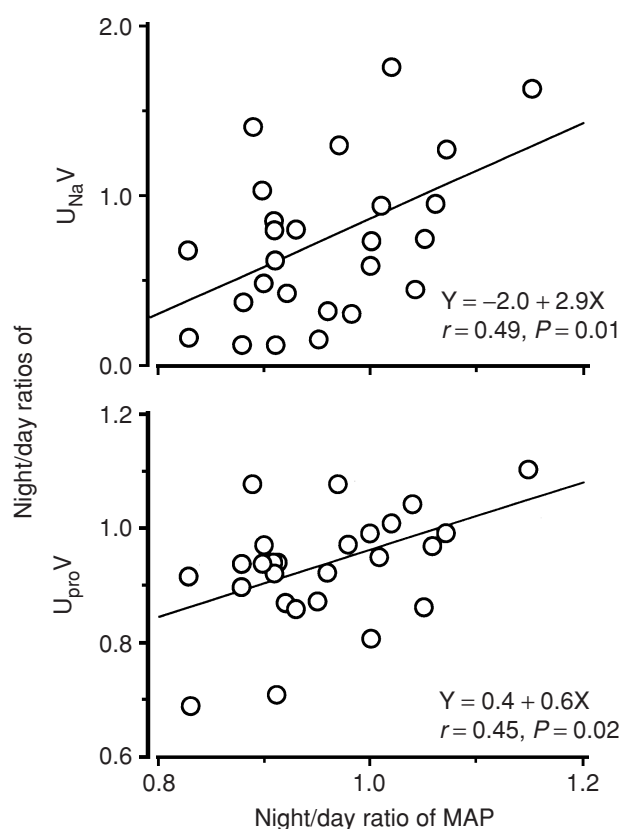


Fig. 2. Relationship of night/day ratio of mean arterial pressure (MAP) (mm Hg) with night/day ratios of urinary sodium excretion rate (U_{NaV}) (mmol/hour) and proteinuria ($U_{pro V}$) (mg/hour) in glomerulopathy ($N = 26$). Night/day ratio of proteinuria was calculated after logarithmic transformation. Night/day ratio of proteinuria itself without logarithmic transformation was also correlated with creatinine clearance ($r = 0.49$, $P = 0.01$).

night/day ratio of MAP and night/day ratio of natriuresis. These ideas are compatible with high sodium sensitivity of blood pressure in glomerulopathy, reported even when GFR is maintained relatively normal [24, 25] and becomes even much higher as renal function deteriorates [26, 27].

The present study showed that the diurnal rhythm of proteinuria was also disturbed, in addition to the diurnal rhythms of blood pressure and natriuresis in glomerulopathy. The diurnal rhythms of U_{NaV} and proteinuria may depend on alternations in the filtered load from glomerulus and/or tubular handlings. Significant positive correlations between night/day ratios of these parameters and night/day ratio of MAP suggest filtered rates of sodium and protein from glomerulus play a key role in creating the circadian rhythms. Albuminuria has been considered as a marker of glomerular capillary hypertension [22]. In fact, it is reported that the elevated glomerular capillary pressure, one of the major determinants of GFR, enhances the transglomerular albumin passage in animal models with experimental glomerulopathies

[21]. These findings suggest that blood pressure change was transmitted to glomerular capillaries, and elevated glomerular pressure contributes, at least in part, to enhanced nocturnal proteinuria in glomerulopathy.

In our previous study with essential hypertension [17], creatinine excretion rate was decreased from day to night, and night/day ratio of creatinine excretion rate was correlated with night/day ratio of MAP. It is well known in normal subjects that GFR is reduced during the night [17, 28], resulting in reduction in creatinine excretion rate [17]. In the present study with glomerulopathy, however, creatinine excretion rate was not different between day and night, and no correlation was found between night/day ratio of creatinine excretion rate and night/day ratio of MAP. In addition, there was no tendency at all that the night/day ratio of creatinine excretion rate was elevated as renal function deteriorated. Although creatinine clearance or glomerular filtration was not measured to compare between day and night, our results suggest that GFR was not reduced during night even when renal function was maintained relatively normal in glomerulopathy. As seen in an early stage of experimental glomerulonephritis [29–31], probably glomerular capillary pressure is elevated to compensate for reduced glomerular filtration capability even when GFR remains relatively normal. Since glomerular capillary pressure becomes dependent on systemic blood pressure in renal dysfunction [32], GFR, and creatinine excretion rate were expected to be correlated with blood pressure changes from day to night, as recognized in proteinuria. The discrepancy between GFR and proteinuria on blood pressure dependency, recognized in the present study, cannot be explained for easily at present.

Nondipper type of circadian blood pressure rhythm is now considered to be a risk factor for the progression of nephropathy [10, 15, 16]. Since degree of nondipping was closely correlated with degree of renal function loss; however, nondipping might be consequently correlated with progression of nephropathy. Our results suggest that it might be a phenotype of impaired renal functional reserve and renal dysfunction. On the other hand, it is well known that proteinuria is also a risk factor for the progression [33]. Based on our findings, proteinuria during night seemed a function of non-dipping of blood pressure. Therefore, if we can normalize the circadian rhythm of blood pressure from nondipper to dipper, proteinuria may be reduced, resulting in retarding the progression of nephropathy. Thus, nondipper pattern of blood pressure rhythm per se also might contribute to progressive renal injury through proteinuria. We have shown that sodium restriction [5] and diuretics [34] can normalize blood pressure rhythm from nondipper to dipper patterns. It will be important to examine whether the measures to normalize circadian blood pressure rhythm, such as sodium restriction and diuretics, can reduce

proteinuria and retard the progression of nephropathy. Since our study was cross-sectional, anyway, prospective studies are required to examine which antedates, loss of renal functional reserve and renal dysfunction or nondipper type of circadian rhythm of blood pressure. Our study does not preclude recently proposal [15] that lack of nocturnal blood pressure fall may antedates both loss of kidney function and enhanced urinary excretions of sodium and protein during night.

CONCLUSION

The present study demonstrated that in patients with glomerulopathy, circadian rhythms of natriuresis and proteinuria were also disturbed in addition to blood pressure rhythm. The degree of the disarrangement was enhanced as renal function deteriorated. Precise mechanisms how such disarrangement occurs remain unsolved.

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